AI-Driven Portfolio Optimization Techniques for Constructing a Smart Index Fund to track S&P 100

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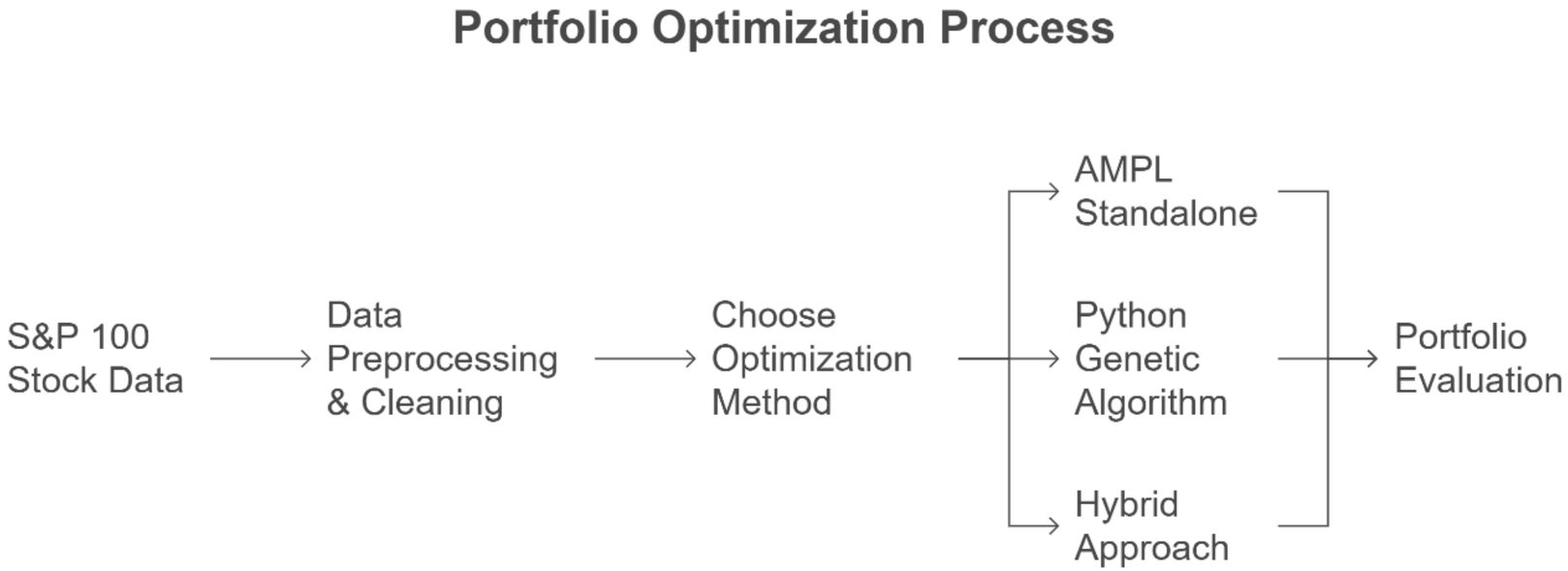
Introduction

Traditional index funds track indices like the S&P 100 using all constituent stocks.

This project aims to create a smart index fund reducing number of assets (q<100) stocks while preserving performance of ETF OEF.

Compare three optimization approaches - AMPL, Python GA, Hybrid to minimize tracking error and evaluate performance.



Problem Statement

- ***Goal:*** Mimic the performance of the S&P 100 using only q < 100 stocks.

- ***Objective:*** Minimize tracking error between the constructed portfolio and benchmark.

$$\min_{\mathbf{w},\mathbf{x}} \quad \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(\sum_{i\in S} x_i \cdot w_i \cdot r_i(t) - R_b(t)\right)^2}$$

- T: number of trading days
- S: full set of S&P 100 stocks
- $S_q \subset S$: subset of selected stocks, where $|S_q| = q$
- $r_i(t)$: return of stock $i \in S_q$ on day t
- $R_b(t)$: benchmark ETF return on day t
- w_i: weight of stock $i \in S_q$
- $R_p(t) = \sum_{i\in S_q} w_i \cdot r_i(t)$: portfolio return on day t

Methodology

Step 1: Data Collection (Yahoo Finance API)

Step 2: Preprocessing (returns, alignment, cleaning)

Step 3: Optimization using 3 techniques

Step 4: Evaluation using metrics and visualizations

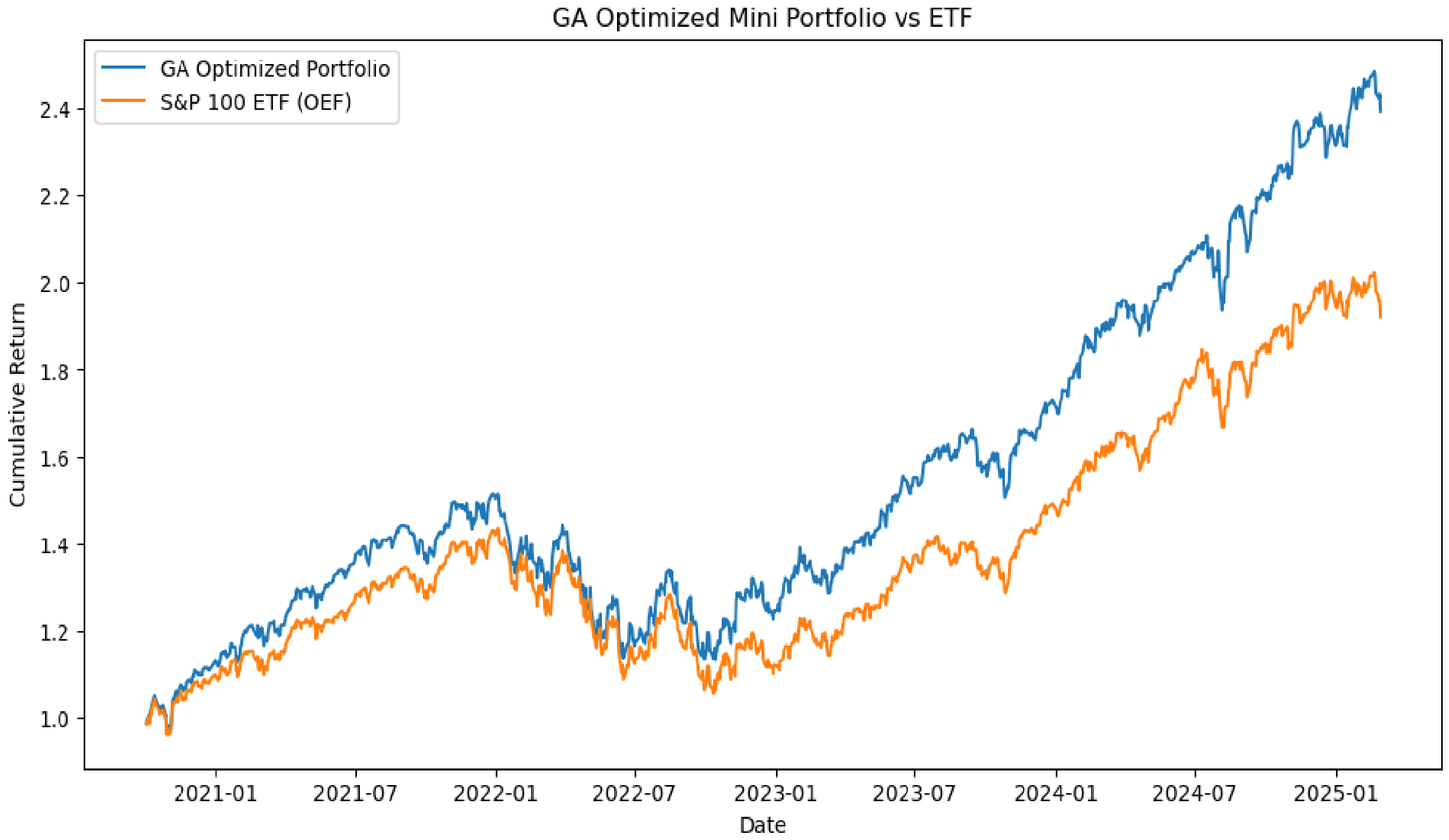
Approach 1: Python-Based Genetic Algorithm

- *Tools:* Python, DEAP, NumPy, Pandas
- *Strategy:* 20 selected stock indices + weights
- *Output:*

✓ Tracking Error: 0.0034

✓ Correlation: 0.9487

✓ Final Cumulative Return: 1.9144



Approach 2: AMPL Standalone Optimization

- AMPL model with CPLEX solver
- *Input:* .mod and .dat files with constraints and returns.
- *Output:*

➢ Tracking Error: 0.0026

➢ Correlation: 0.9718

➢ Final Cumulative Return: 2.6031

```
PS C:\Users\moham\AMPL\ampl> ampl
ampl: reset;
ampl: option solver cplex;
ampl: include run_index.run;
CPLEX 22.1.1: optimal solution; objective 0.007438187292
0 simplex iterations
w [*] :=
 AAPL  0.148735
  ABT  0.0325095
 AVGO  0.0610043
  AXP  0.0354142
 BKNG  0.0139669
BRK-B  0.109884
    C  0.0569377
   CL  0.0595313
 GOOG  0.0697454
GOOGL  0.0445109
 INTU  0.022565
  LIN  0.0260445
  MCD  0.0629184
   MO  0.0407181
  MRK  0.0471924
 NFLX  0.0256679
  NOW  0.033732
 NVDA  0.0602021
 TMUS  0.0299315
  UNP  0.0187891
;
```

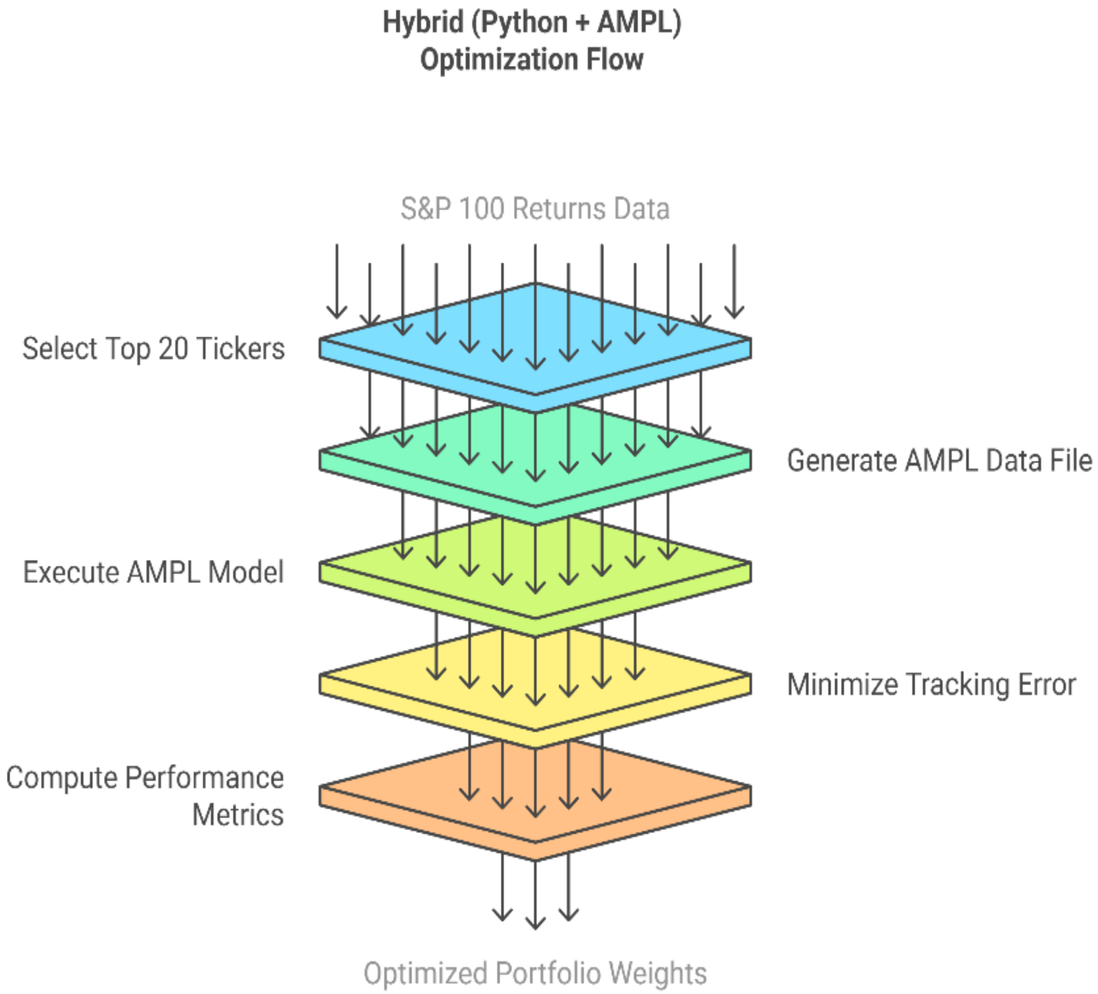
Approach 3: Hybrid Python + AMPL

- Python generates .dat file, calls AMPL via subprocess.
- AMPL performs optimization & results read back by Python.
- Combines data handling and solving power.
- *Output:*

➢ Tracking Error: 0.0026

➢ Correlation: 0.9718

➢ Final Cumulative Return: 2.6031



Performance Comparison

All three methods use the same dataset and objective function

Python GA:
Tracking Error = 0.0034
Correlation = 94.87%
Return = 1.91x

AMPL:
Tracking Error = 0.0026
Correlation = 97.18%
Return = 2.60x

Hybrid:

Tracking Error = 0.0026
Correlation = 97.18%
Return = 2.60x

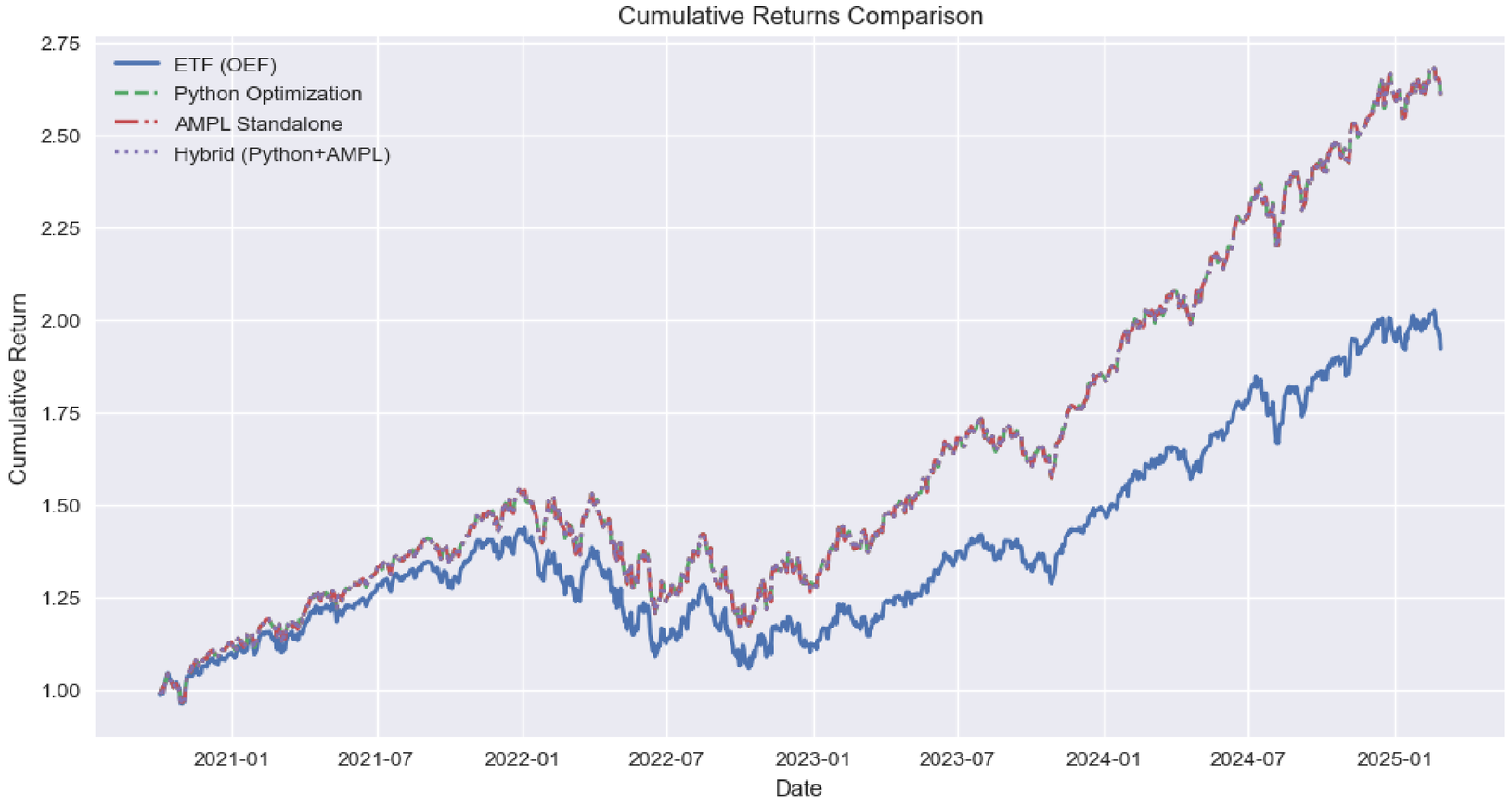
q=20 found optimal

All 3 approaches perform equally well.

	Tracking Error	Correlation	Final Cumulative Return
Python Optimization	0.0034	0.9487	1.9144
AMPL Standalone	0.0026	0.9718	2.6031
Hybrid (Python+AMPL)	0.0026	0.9718	2.6031

Insights & Observations

- Increasing q reduces TE but 20 is a practical trade-off.
- All methods show similar performance due to shared constraints.
- GA allows flexibility, AMPL is precise, Hybrid combines both.



Conclusion & Future Work

- Optimized portfolio of 20 stocks closely tracks S&P 100.
- *Python GA:* Flexible, exploratory.
- *AMPL:* Robust and fast.
- *Hybrid*: Best of both worlds.
- *Future:* Sharpe Ratio, Sector Constraints, Reinforcement Learning.